

Unpacking the internal assessment

Criterion D: Development

Functionality

Functionality refers to how well the product works. The product should address the problem described in criterion A. To show functionality, the video should preferably include a single test-run that shows the performance of the solution following the testing strategies developed in the system overview.

A product that does not address the problem specification (criterion A) cannot achieve more than limited functionality. A product that addresses an inadequate or vague problem specification will only achieve partial functionality. Full functionality can be achieved if the product is a working solution to an adequate problem specification, even if the success criteria have not been fully met.

Functionality will be assessed on the quality of the test-runs in the video and on the evidence of testing included for criterion D.

Algorithms and techniques

The process of turning algorithms into code is a key element in developing a product that solves the stated problem. This process will require students to use different programming techniques. It is essential that they clearly demonstrate these techniques in writing and by showing excerpts of code. Students should provide comments with the code to explain its purpose.

This criterion gives students the opportunity to show their knowledge and understanding of tools and techniques, which must be explained sufficiently to fulfil the intention of “development”. There must be explicit evidence of algorithmic thinking in the form of code excerpts, along with explanations and justifications of the chosen techniques.

It is also important that students discuss and evaluate the choices they made when deciding which techniques were most appropriate.

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It is acceptable to improve code with a large language model (LLM) such as ChatGPT, but any code, content or ideas generated by the LLM must be acknowledged. Students must include a comment in their code and a note in their documentation indicating where and how they used the LLM.

Student example 1:

"I wrote a function to sort a list, but it had a logic error. ChatGPT suggested using Python's `sorted()` function, improving efficiency from $O(n^2)$ to $O(n \log n)$. I've acknowledged this change in my comments."

Student example 2:

"I needed to write a function to calculate the Fibonacci sequence, but I wasn't sure how to implement it. I asked ChatGPT to generate the code, and it provided a recursive solution. I used this code in my project and added comments to indicate that ChatGPT generated it and to explain how the algorithm works."

Note that documenting the development should **not** be a process journal. Instead, the student must highlight, explain and justify specific tools and techniques used.

Effective testing strategies

The effectiveness of the testing strategies used is documented and discussed in this section. Evidence of the implementation of at least three separate test cases must be included, showing both functional and structural testing.

This could be addressed by copying the test tables for a few major algorithms from the testing strategies section in the system overview, adding a new column for the observed results. These results should address valid, extreme and invalid data. It is expected that students will refer to evidence of testing seen in the test-runs in the video.

Students can justify the effectiveness of testing strategies by evaluating the comprehensiveness of the testing.

